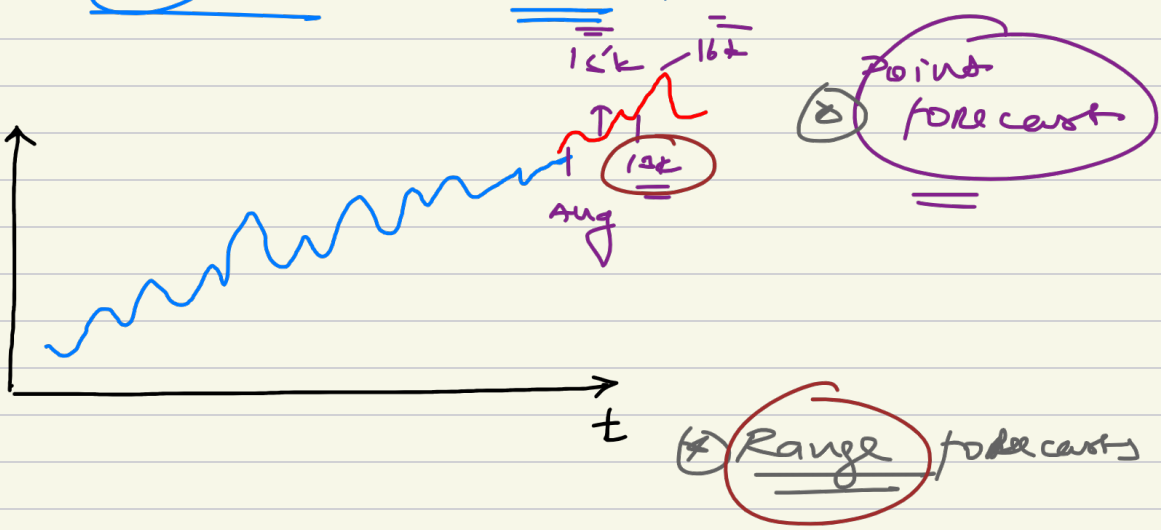
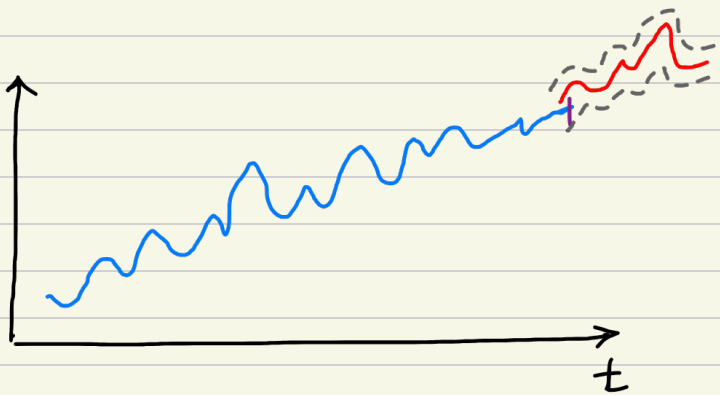
Time Series Analysis - 4

SARIMA

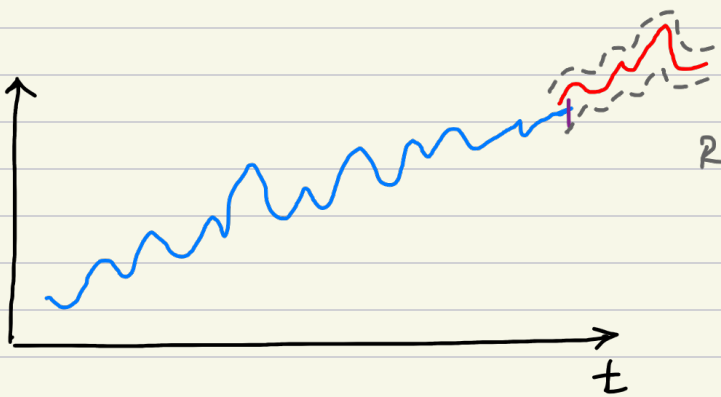
- Agenda ⇒ Confidence intervals (Range forecast)
  - Change Points ⇒
  - Exogeneous variables
  - SARIMAX → visitors

Time series → Sales, Visitors, Calories





Confidence intervals



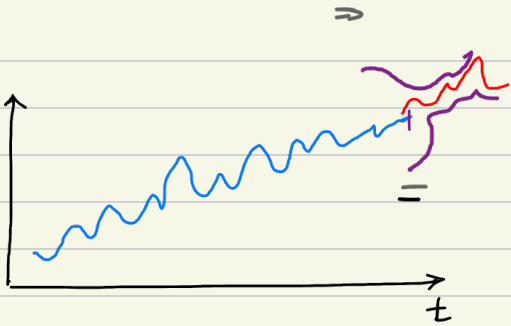
Range forecast.

Point → 250 ⇒ Product

Range → [235 — 265] → 95% confidence

250 ± 15 units





1 ✓

2 Practical

Reasoning

Real world

Town

Weather

EOD

Next day

Electricity

Next year

⇒

Time passes → , uncertainty ↑

Range →

r. age

↳ confidence

Confidence interval

90% → [280 – 320] units

95% [260 – 340]

99% [240 – 360]

95% interval forecast

Model ⇒ point → Sales 100 lacs → Best 100 ✗

Range → 80 lac – 120 lac

Repeatedly × 95% confidence

PF

Temperatures

Point = 30°C

95% cont. interval

24°C — 36°C

30°C

Range

1000 units

Business — 1000 ✓

95% cont.

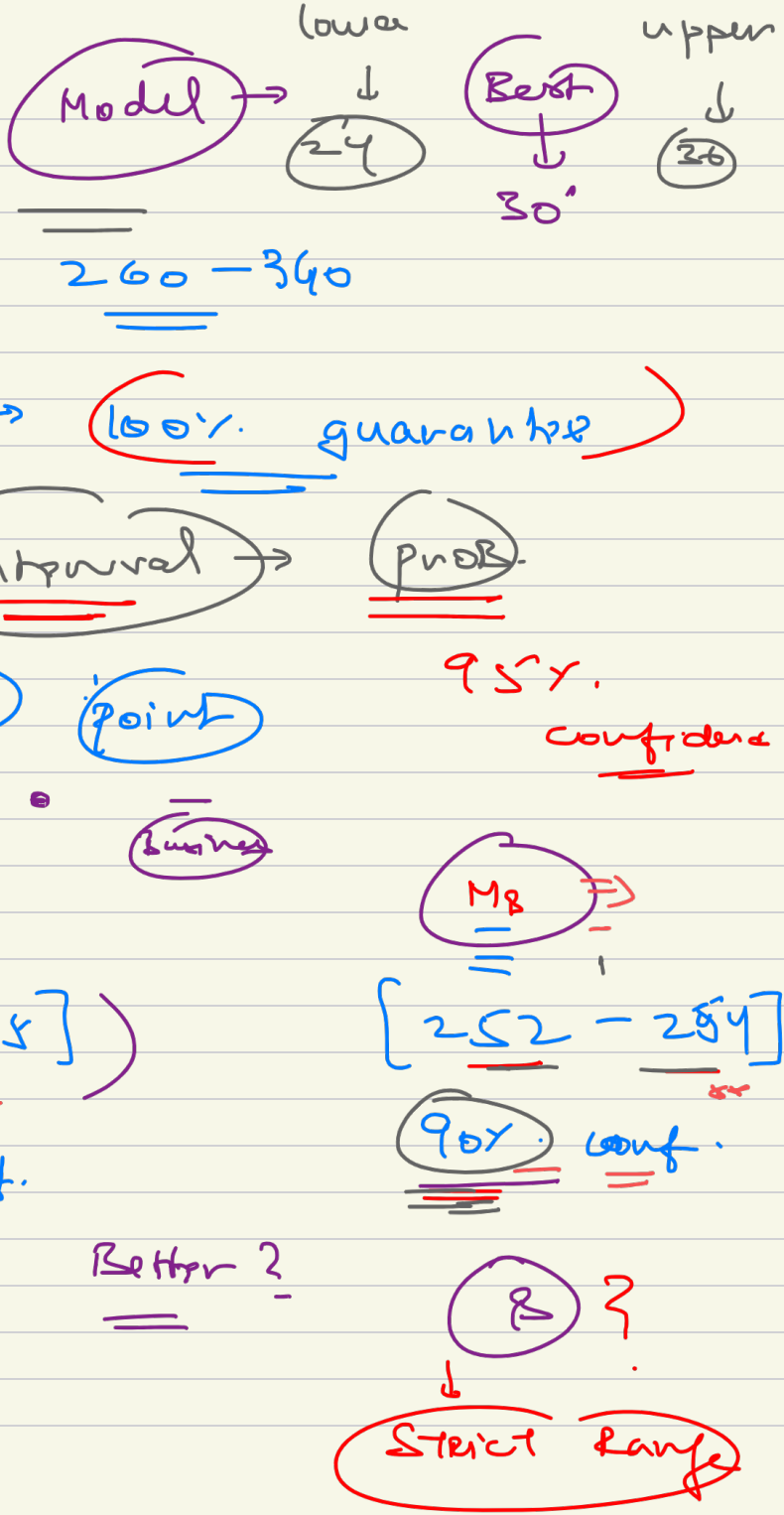
(800 — 1200

Range

forecast = 1000

lower = 800

upper = 1200 ⟹


Model
lower
24
Best
30'
upper
36
260 - 340
Range
100% guarantee
interval
prob.
95%.
confidence
forecast
point
M_A
[250 - 255]
90% conf.
M_B
[252 - 254]
90% conf.
Better ?
B ?
Strict Range

Indians → gay

100 - 250 *

180 - 195 *

A

M_B B

95%. ⟹

([250 - 255])

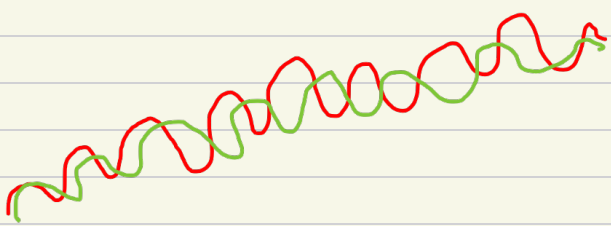
M_A
↓
90%

M_B ✓

↓
95%.

Calculating Ci

training data
=



AR, MA
ARMA
=
↓
pred
=

y — actual

$\hat{y}$ — pred^{n}

Residuals

$y - \hat{y}$



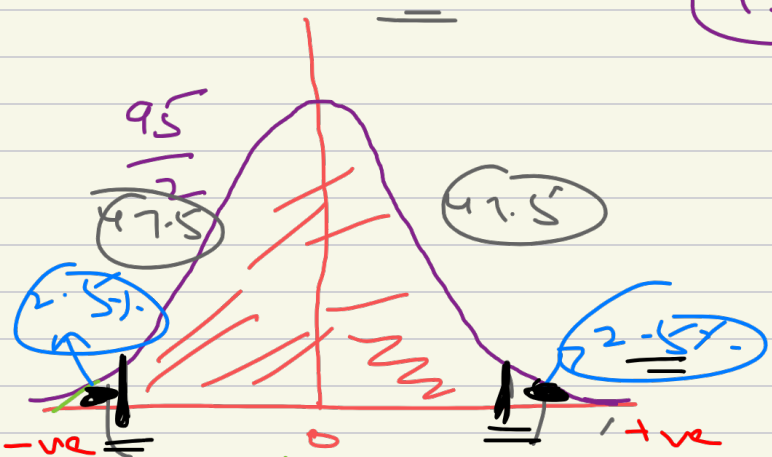
(*) dist^{n} / hist / KDE

Normal dist^{n}

95%

$\frac{95}{2} = 47.5$

z score



cdf, PPF

CI

Range estimate

± 7

95% confidence

$\hat{y} = 250 \pm 7$

$[243, 257]$

95% confidence

99%

99/2

99/2

0.5

0.5

50%

50%

AR, MA

cdf ppf

Statistical

$250 \pm ?$

10:10

Hospital staffing

Patients = 300 ↑

95% → [270 – 340]

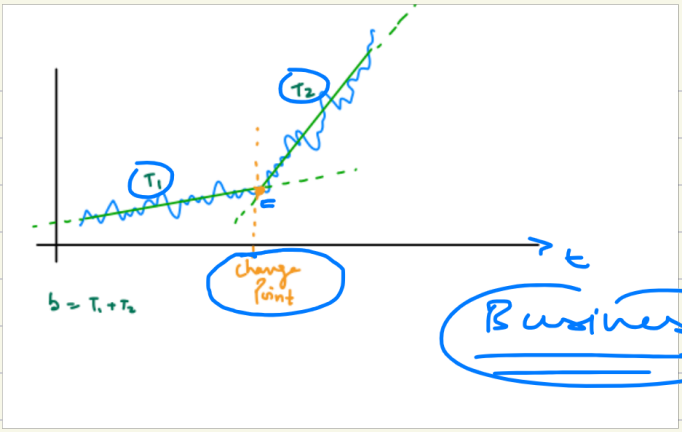
~~expected = 300~~

95% → 180 – 450

Range forecasts

Change Point → not used for forecasting

↳ analysis

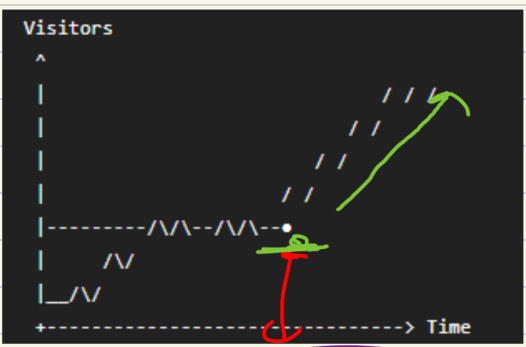


Blue → T3

Business

100 102 98
97

150, 153 IP



Statistical Behaviour

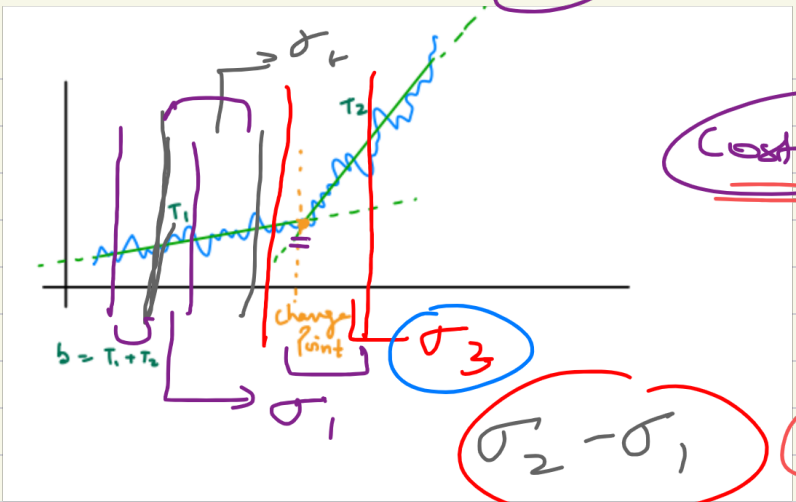
Change time

(3) (what exactly can change)

(1) Level

(2) Trend (slope)

(3) σ^2 or σ



T_3

algo

Cost func.

μ, σ, slope, ...

$\sigma_2 - \sigma_1$ σ

change → v-small

$\sigma_3 - \sigma_2$ →

Shift the window keep computing the cost

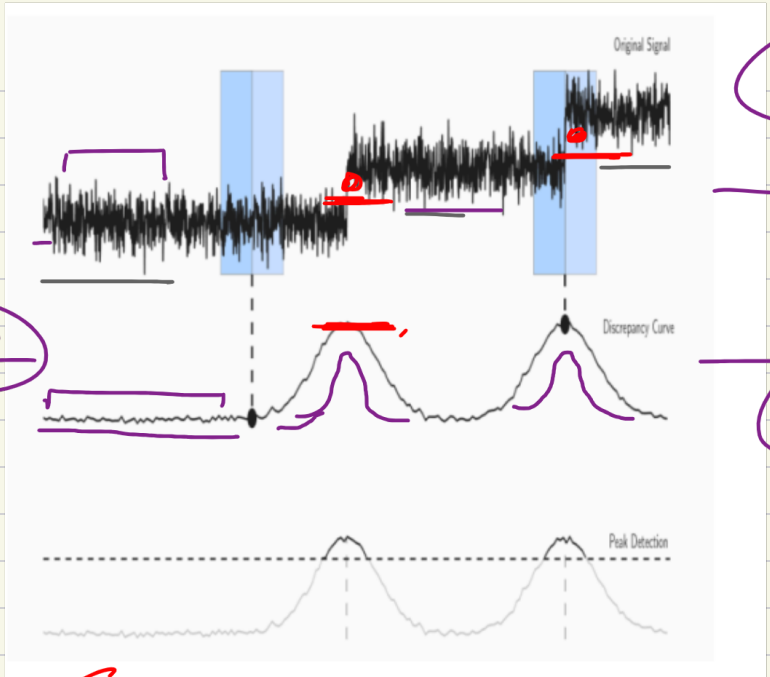
How

$|c_p - c_{i+1}| > \xi$

σ_K

Change points

cost = σ



Ci – Ci'

→ original

cost

cost time

why ?

movie Tickets

2018 → 2019 → early 2020

change point

problem

covid

↓

STRUCTURAL BREAK

Demonetization lockdown produces largest coupets dor

Flipkart ⇒

Jan – Jul

20k, 21k, 20k, 22k

Aug ↓

22k, 21k, - - -

Sept →

35k, 37k, 40k ⇒

Flipkart

Same day delivery

(lower)

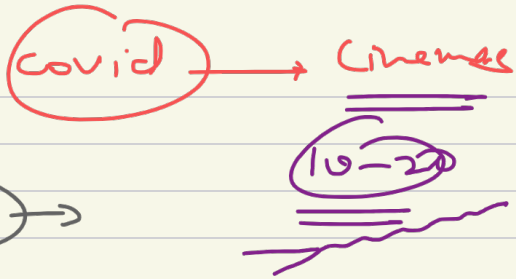
Sept → change point

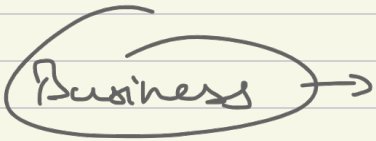
TS → Business

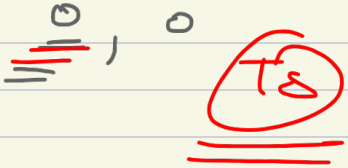
Specific

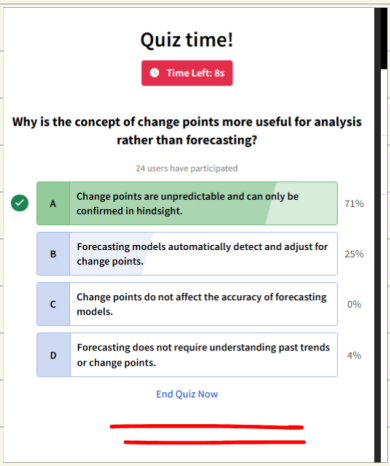
University



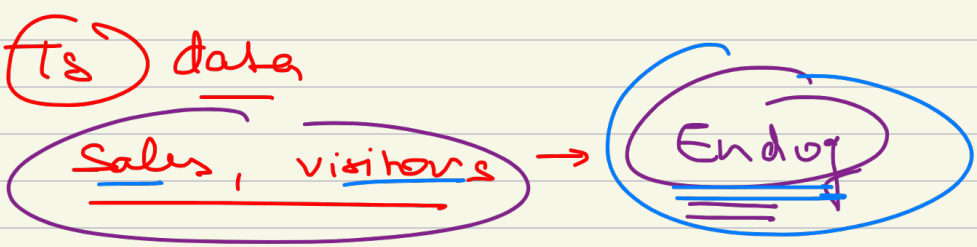


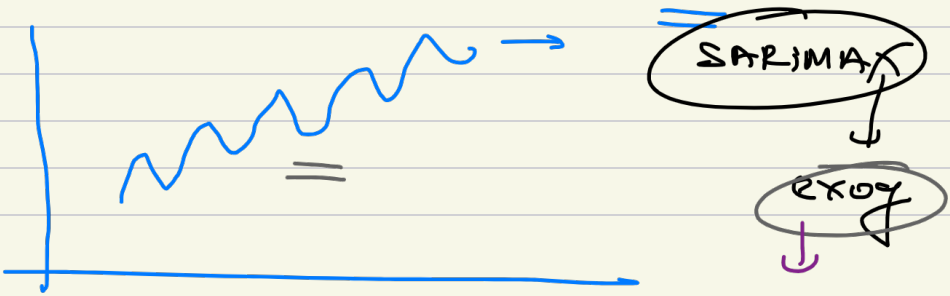






Exogenous VAR





(external info)

Visitors

ARIMA | SARIMA

Holiday Today?

Marketing campaign

Temperature?

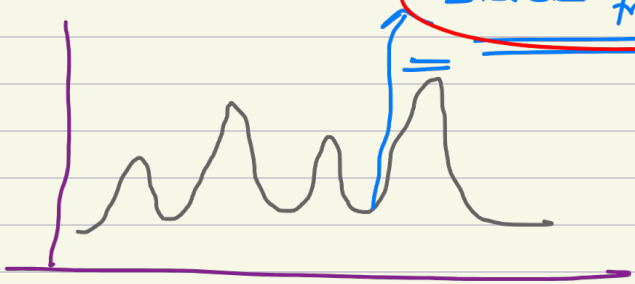
Rainfall

→ influence customers

Error → error of info

SARIMA

Black friday ⇒



Endog

Visitors
Sales
Demand
Stock qty

Exog

- Holiday
- Temp
- Adv.
- Rainfall

apple → external

i

Apr → 0
Aug → 0

Sept → 1

X

✓

iphone 17

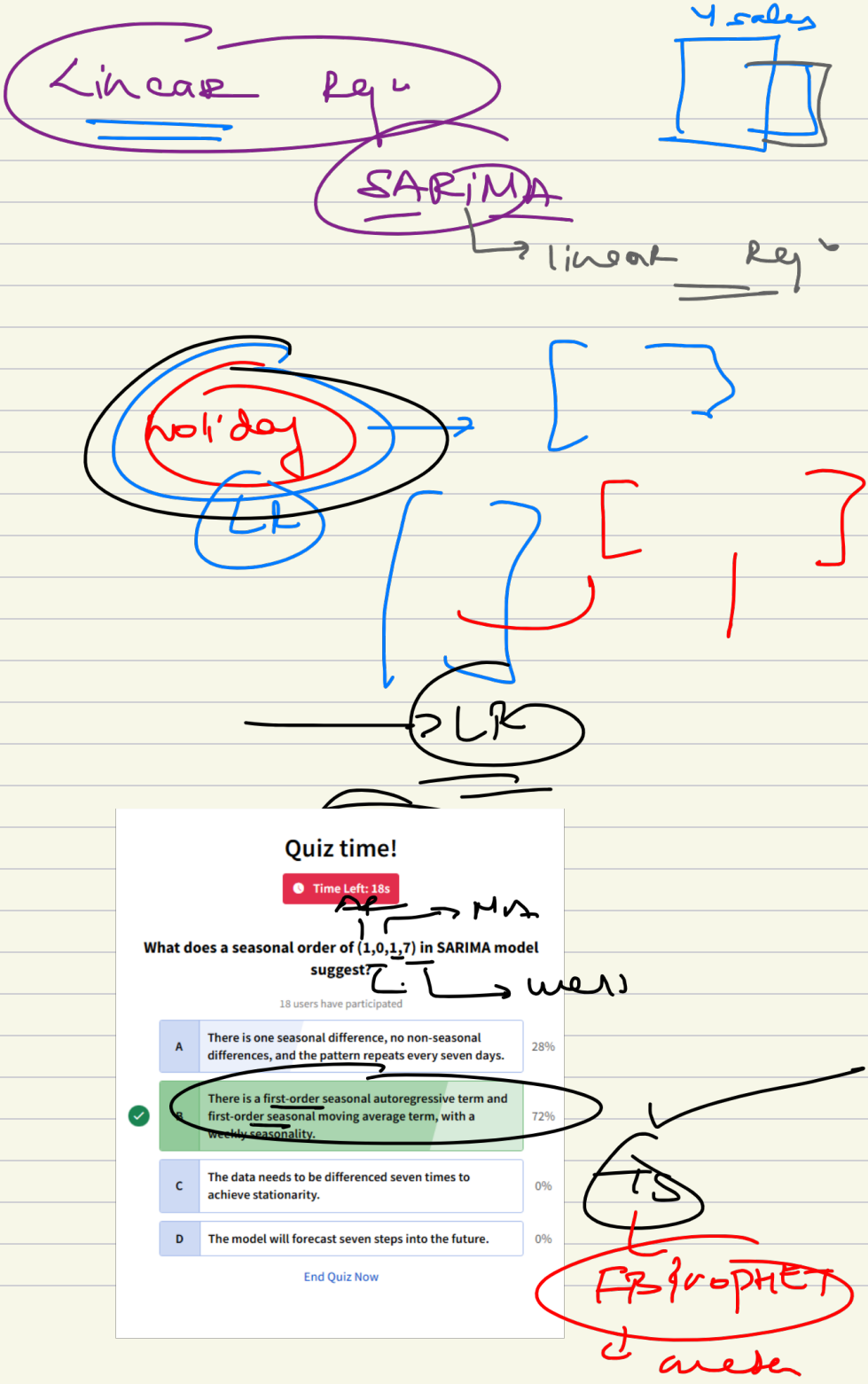
15 | 16

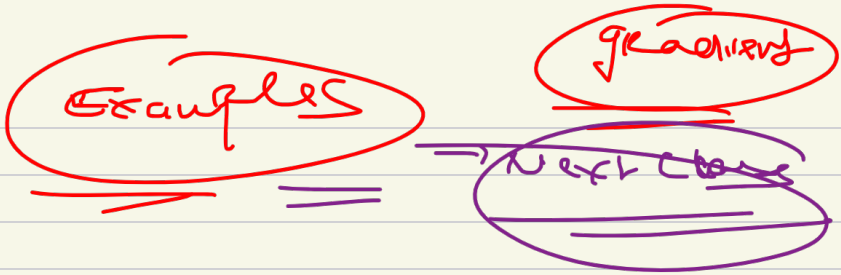
i/p

$$y(t) = b(t) + s(t) + e(t)$$

holiday

$$y(t) = b(t) + s(t) + h(t) + e(t)$$


Linear Reg
SARIMA
linear Reg
holiday
LR
LR
Quiz time!
Time Left: 18s
What does a seasonal order of (1,0,1,7) in SARIMA model suggest?
18 users have participated
A There is one seasonal difference, no non-seasonal differences, and the pattern repeats every seven days. 28%
B There is a first-order seasonal autoregressive term and first-order seasonal moving average term, with a weekly seasonality. 72%
C The data needs to be differenced seven times to achieve stationarity. 0%
D The model will forecast seven steps into the future. 0%
End Quiz Now
TS
PROPHET


Examples
Gradients
Next Class